

Inductor Analysis Flow

1. Minimum Frequency Analysis

Analysis: AC at POI 6 GHz

OUTPUT	EXPRESSION
V_L	$V(/in_p)$
$L_{\max}$	$\text{imag}(V_L) (2*3.1416*6e9)$
Q	$\text{imag}(V_L)/\text{real}(V_L)$

2. Maximum Frequency Analysis

Analysis: AC at POI 9.5 GHz

OUTPUT	EXPRESSION
V_L	$V(/in_p)$
$L_{\min}$	$\text{imag}(V_L) (2*3.1416*9.5e9)$
Q	$\text{imag}(V_L)/\text{real}(V_L)$

3. Corners

Setup: Temperature_Process

The analysis includes sweeping across the following environmental and manufacturing conditions:

- **Temperatures:** 11 discrete temperatures ranging from -40°C to 125°C .
- **Processes:** 3 tandem corner processes (SS, TT, and FF).

In this case, each inductor model is a combination of a process and a temperature, yielding a total of **33 corners** (11 temperatures $\times$ 3 processes).